

Predicting and controlling the growth shape of crystals – A solvent selection approach

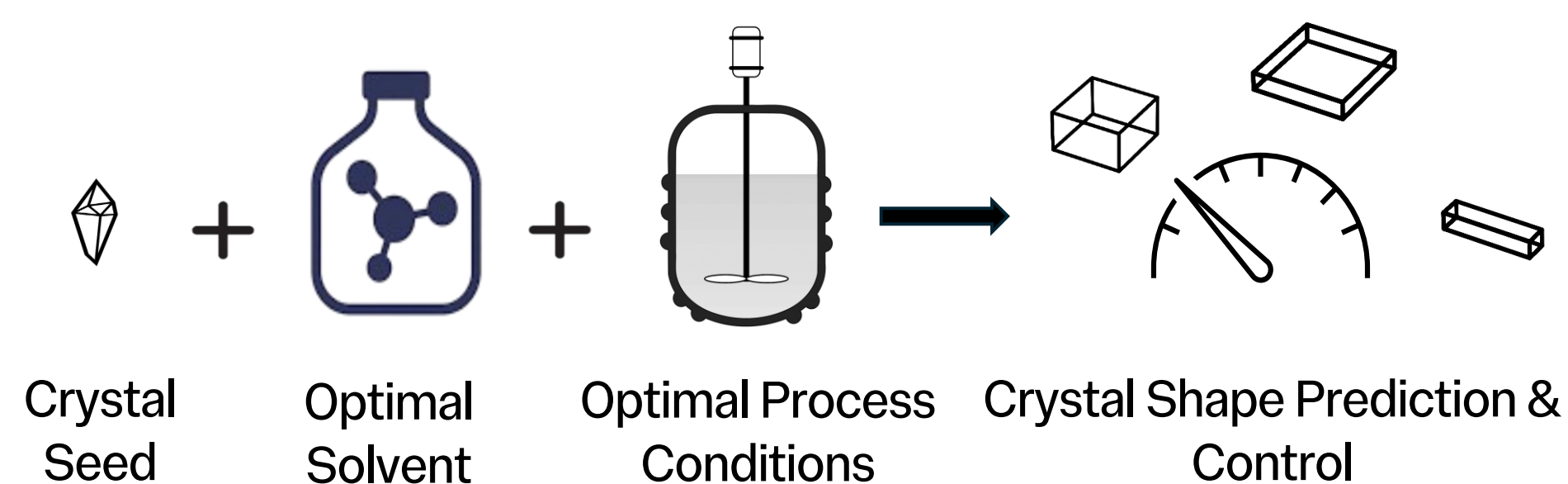
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Introduction

Problem Needle shaped crystals clog filters, difficult to handle

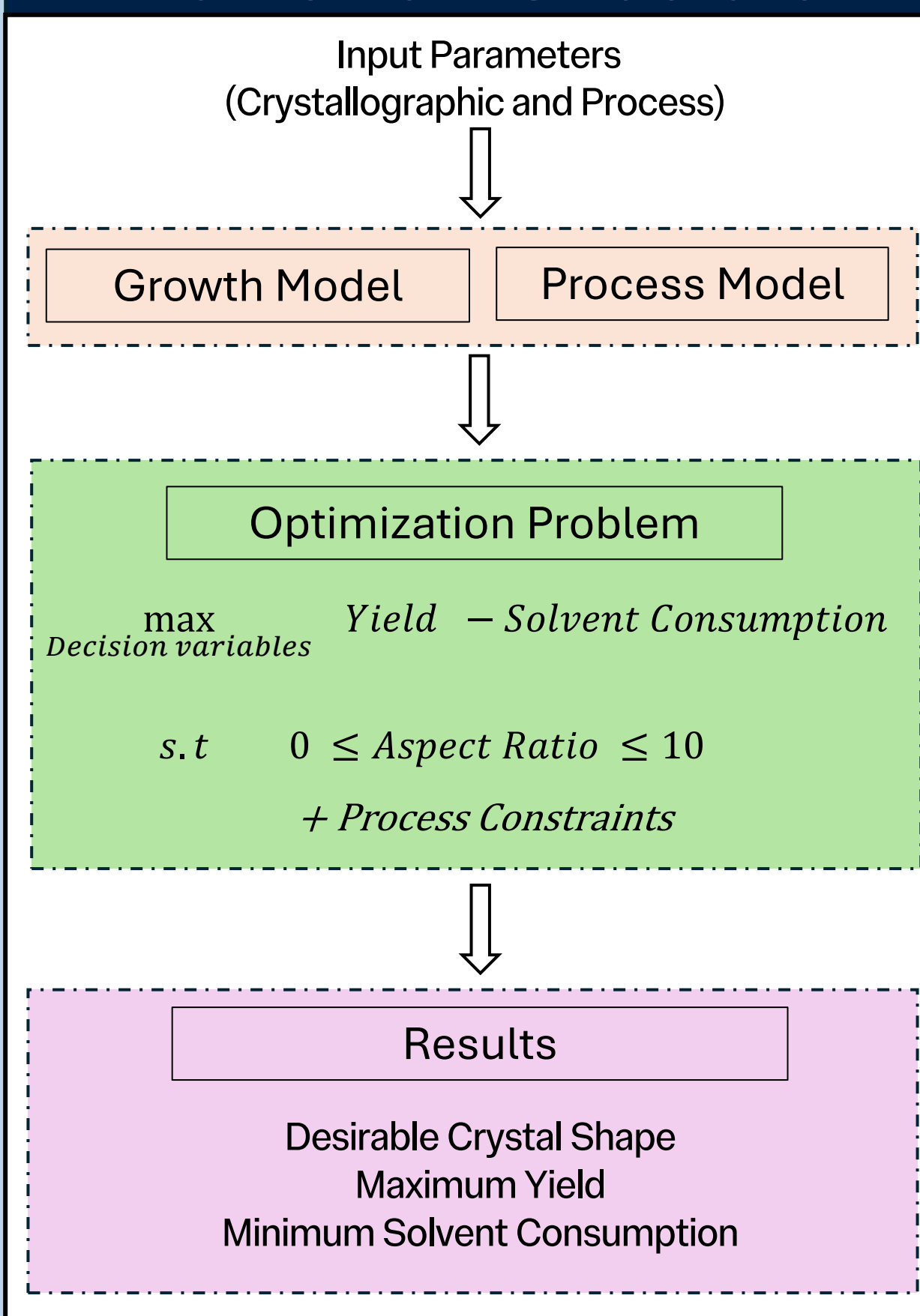
Solution Avoid undesirable crystal shape formation



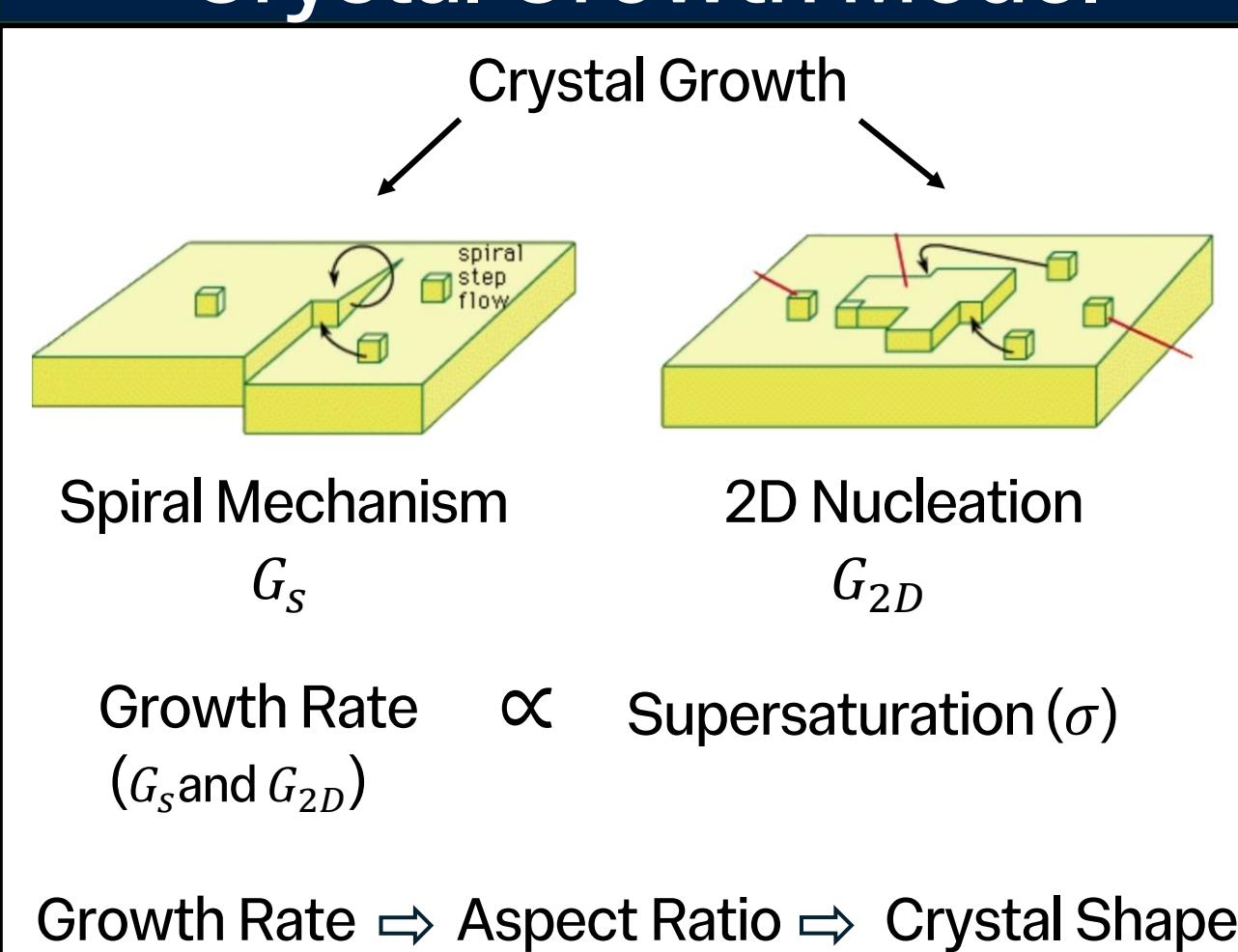
Research Aims

- Develop integrated modelling framework combining crystal growth model and process model.
- Predict crystal shapes in different solvents and at different temperatures
- Using modelling framework, identify optimal solvents and process conditions to maximize crystal yield and minimize solvent consumption.
- Simulate the complete cooling crystallization process.
- Observe evolution crystal shape in cooling crystallization process

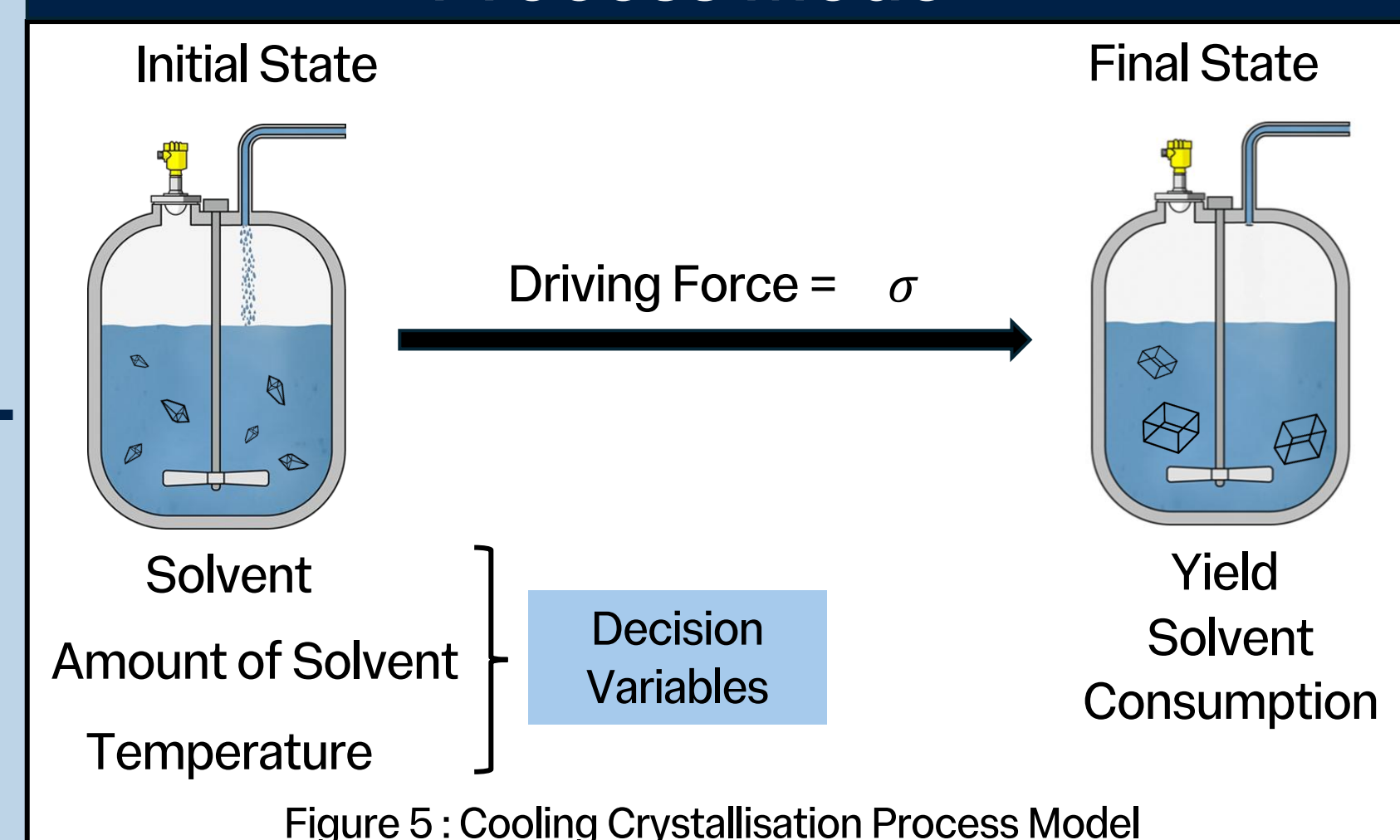
Framework Structure



Crystal Growth Model



Process Model



Results - Optimization

- Optimal solvents and process conditions were identified for Biphenyl and Naphthalene Crystallization

Biphenyl Crystallization

Rank	T (K)	Optimal Solvent	Amount of Solvent added (Moles)	Aspect Ratio	Yield* (%)	Solvent Consumption (g/g)
1	308	Toluene	1.743	9.97	1.02	101.33
2		Butyl Acetate	4.686	9.8	0.78	451.04
3		Hexane	14.123	5.18	0.7	1124.6

Naphthalene Crystallization

Rank	T (K)	Optimal Solvent	Amount of solvent added (Moles)	Aspect Ratio	Yield (%)	Solvent Consumption (g/g)
1	306	Acetone	112.362	8.01	32.62	156.05
2		1-Pentanol	360.168	8.02	32.49	762.31
3		Butanol	441.355	8.03	32.48	785.77

Conclusions

- Optimal solvents and process conditions for yield maximization and solvent consumption minimization were identified.
- Trade-Off between Aspect Ratio, yield and solvent consumption is observed.
- Decrease in aspect ratio is observed when system is cooled uniformly

Future Work

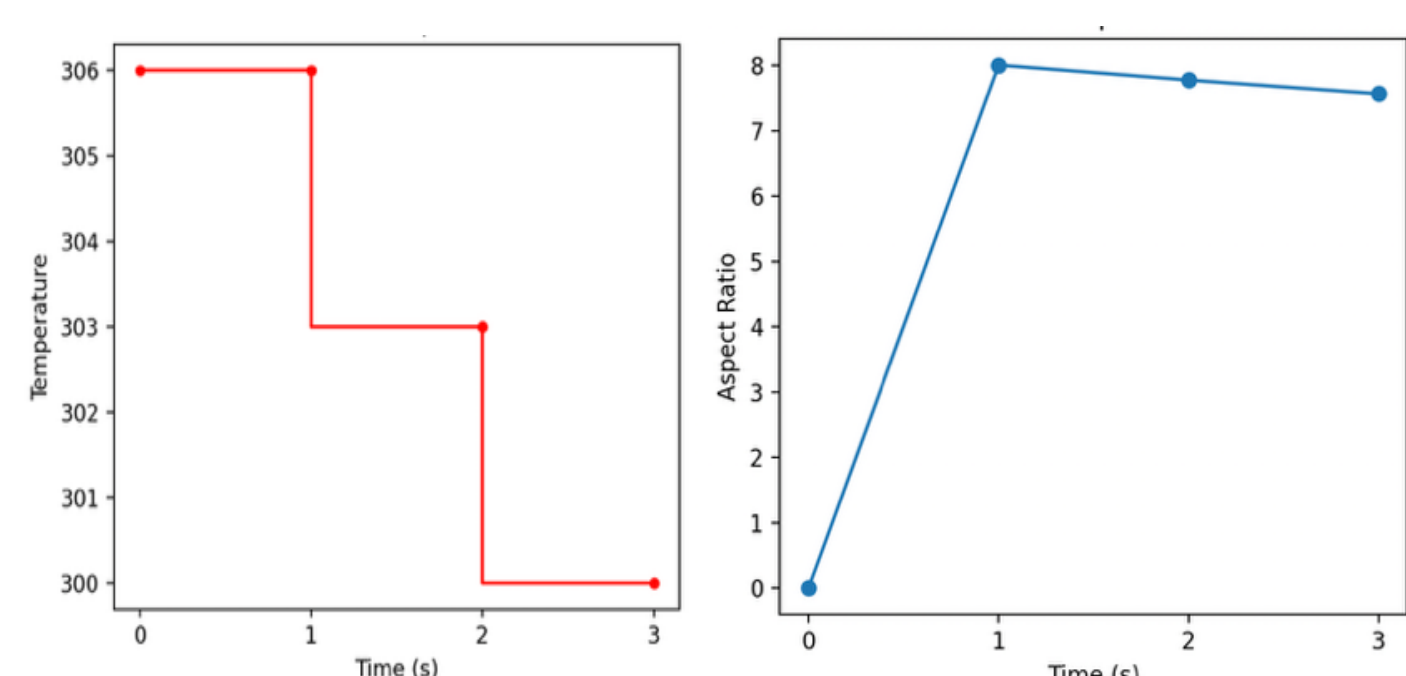
- Improve shape prediction results
- Include rate of cooling as a decision variable
- Include HSE considerations for solvent selection
- Perform economic evaluation and obtain optimal trade off values.

References

- F. Civati et al. *Cryst Growth Des*, vol. 21, no. 6, pp. 3449–3460, Jun. 2021, doi: 10.1021/acs.cgd.1c00217.

Results – Simulation of Cooling Crystallisation Process

Biphenyl Crystallization



Naphthalene Crystallization

